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Predicting Travel Time in Complex Road Structures using Deep Learning

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***Abstract*—Vehicular traffic and congestion is a major challenge worldwide because of rapid growth in urban population. The congestion can be mitigated to enhance traffic management by predicting accurate travel time of the vehicles in the traffic. This research developed a novel methodology utilizing machine learning on real-time traffic data collected through Bluetooth sensors deployed at traffic intersections to estimate travel time. The research evaluates performance and accuracy of five different prediction systems for travel time estimation highlighting the effectiveness of the machine learning models in accurately predicting travel time. The research also explores the development of the machine learning model predicting the travel time during peak hours, considering traffic lights impact on travel time between intersections. This research findings contribute to the efficient and reliable travel time prediction systems development, helping commuters making informed decisions and improve traffic management strategies.**

***Index Terms*—Travel time, LSTM, Neural Network, Bluetooth sensor data, one hot encoding**

I. Introduction

Travel Time prediction is a fundamental concept for knowing the travel information ahead. It is defined as the amount of time consumed by a moving vehicle to move from one intersection to another. The ground truth travel time has been defined in many different ways [1] . Travel Time plays an important role for measuring traffic efficiency and for estimating the delay on the road. For us to predict the travel time in a real world situation it is important for us to get the necessary data for predictions in the same location where we want to predict the travel time [2]. It is not possible that the same methodology can be applicable to other locations as the traffic density might be very high which gives inaccurate results for predictions.

Now a days, Bluetooth detectors were emerged and are introduced in the current research and were configured in the research lab in order to track the movements of the vehicles between intersections [3]. The evolution of the road traffic flow will have a very complex structure as the traffic density is increasing day by day and also there are many new ways that are introduced to mitigate congestion.

In the earlier days, the Travel Time is calculated as the average travel time of each vehicle take to travel from one intersection to another [5] . Assume there are 'n' number of vehicles, then the predicted time of next individual vehicle would be calculated as P = 1/n(sum of all travel times) . This prediction will also work sometimes when there is a constant flow of traffic between two intersections. In the recent years, when we consider the real world scenario the travel time needs to be calculating keeping in mind all the situations such as which day, which time of the day and how many number of vehicles are passing between two intersections.

The main reason why this approach is different from other Google Maps/ Bing Maps/ Apple Maps is that, all the predictions that are made by mapping services are only using the current GPS mechanism and a lot of them will try to find a group of devices in a given radius and will determine if that particular road is busy or not and will suggest a new route if there is any present which can take less time when compared to the current route [4]. But as the one and only limitation for these kinds of approaches would be when the signal lights turn red. There will be delay added to the current ETA and we will have a new ETA that is shown. To reduce the delays and to increase the trust in travel time prediction we generally extract the data from traffic controllers and apply machine learning models on it for predictions so that, we will have a better take on it when compared to the normal

II. DATA COLLECTION AND METHODOLOGY

There has been a significant increase in traffic density, which led to the rise of challenges in congestion management. Estimating the travel time accurately is important to help the commuters during peak hours and traffic controllers to act timely according to the live traffic conditions [6]. The current research proposes a new methodology which implements a machine learning model on real time traffic data collected through Bluetooth sensors at traffic intersections to mitigate congestion.

The data in this paper has been collected in the section of highways in Texas (TX) state, specifically in Austin, with the help of Texas Department of Transportation (TDOT) . According to the topographic environment characteristics and the flow of traffic, TDOT has set up Bluetooth sensors in the traffic controllers to detect the vehicles movements between

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two traffic intersections and also a section of traffic flow has been constructed to understand the real-time data procurement such as origin intersection, destination intersection, speed, travel time.

Bluetooth detectors have been installed on different roadways and is detected in real time [7]. The ground truth travel time of a moving vehicle is calculated by subtracting the timestamps of the same Bluetooth address captured at two different travel intersections.

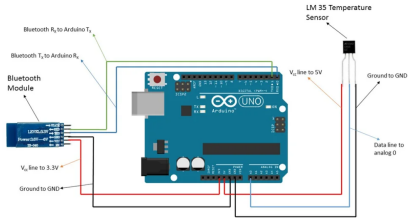


Fig. 1. Bluetooth data Collection

As shown in the above figure the Bluetooth detectors is a point of Bluetooth detectors of traffic information based on a digital radar wave technology. This can obtain information such as sample, average speed of the traffic flow in real time [8]. These Bluetooth detectors are designed in such a way that they generate the data all day long with high accuracy and a very low false detection rate [11]. The basic principle of detection is first assuming that the length of the vehicle in the detection area is first set to a value and then we calculate the driving speed by the difference between the vehicle entering and leaving the detection area. All the vehicles that have the capability of sharing the audio by Bluetooth from mobile devices, iPads, laptops should have a bluetooth receiver set up in built when a car is manufactured. The detectors capture the MAC address of the device which plays a key role for collecting traffic information and differentiating between them.

The above figure shows the configuration of the bluetooth detector and to which HOST IP address the data is being posted.

In this data, we have ten effective fields Record ID, Destination Adddress, Origin Reader Identifier, Destination Reader Identifier,Travel Time (Seconds), Speed (Miles Per Hour), Match Validity, Filter Identifier, Start Time, End Time, Day of Week, with the help of a Bluetooth detector as the basic data of this research.

The original data cannot be directly obtained the ground truth travel time information. The travel time is calculated out by a unique method. Consider there are two points P1 and P2 and the distance between these points is given as d. Then the travel time is extracted by capturing the speed of the vehicle between two traffic intersections and the length

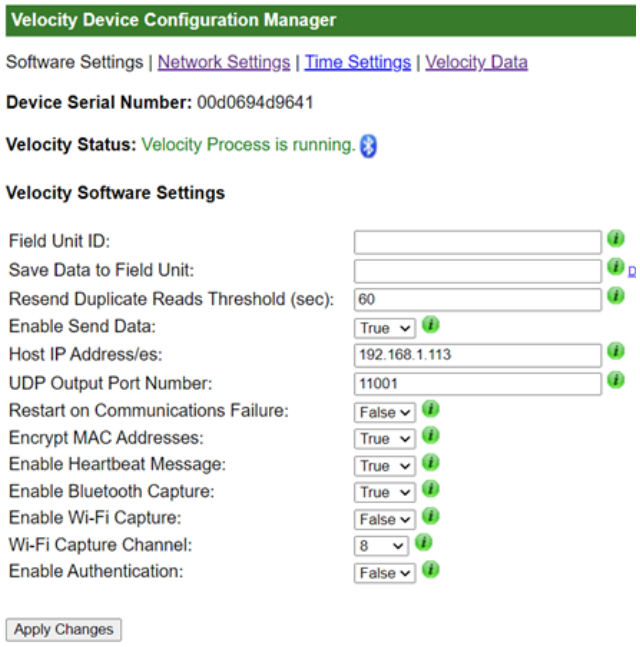


Fig. 2. Bluetooth Detector writing to the Host IP address

of the segment [10].

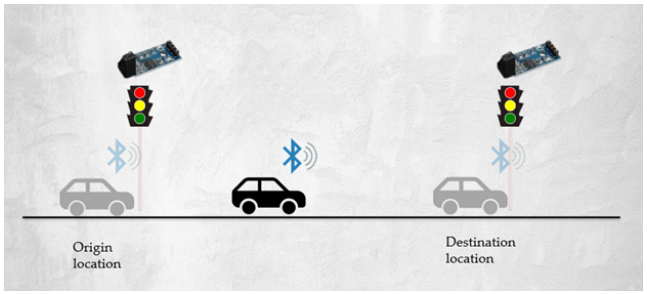


Fig. 3. Bluetooth data Collection

The data in transportation can be collected from various sources such as the data from traffic centers, the data from simulation models, the data that is collected with the intention of doing research by a trusted agency / transportation department [9]. When it comes to simulation models, they require a lot of human effort to generating the data and one of the drawbacks for this kind can be generating the indirect data which cannot be practical and cannot be compared to the real – world data [12]. The data that is obtained from traffic centers which has the control to the overall state is collected in real time and this kind of data needs a lot of pre processing and data cleaning to obtain a good shape to the data. This is the data which is collected by Texas Department of Transportation. A description of the different traffic parameters is as follows:

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Record ID : Record ID is a unique identifier assigned to each record captured by the Bluetooth detector system. It serves as an important identifier for tracking a specific vehicle's journey from the origin to the destination reader.

Device Address : The device Address is the unique MAC address of the Bluetooth device installed on a vehicle. The Bluetooth detector system captures the MAC address of the Bluetooth device and records it along with other parameters for analysis and modeling.

Origin Reader Identifier: The Origin Reader Identifier refers to the Bluetooth detector installed at the intersection where a vehicle first enters the monitored roadway network. This identifier helps in determining the starting point of a vehicle's journey

Destination Reader Identifier: The Destination Reader Identifier is the Bluetooth detector installed at the intersection where a vehicle exits the monitored roadway network. This identifier helps in determining the end point of a vehicle's journey and is critical for calculating travel time and distance between intersections.

Speed (Miles Per Hour): Speed(MPH) is one of the most common parameters for commuters and it has the most significance in determining travel time. Speed is calculated in miles per hour between the origin and destination readers. Consider, a vehicle that travels 5 mph over the given speed limit will reach the destination first when compared to the vehicle moving at an average speed (during peak hours) with the flow of traffic or even if the vehicle is moving at the speed limit, then we can clearly say that the time taken will be less when compared to the first vehicle.

Start Time: Start time is obtained when a vehicle passes through the Bluetooth detector for the first time. We can also say that Time will be important because we can determine if that is a peak hour or not by determining if this is a peak hour or not.

End Time: End Time is the time when a vehicle passes through the Bluetooth detector installed at the Destination Reader Identifier. It is an important parameter for calculating the travel time and understanding the traffic flow on the roadway network.

Day of the Week: Day of Week refers to the day on which a vehicle passes through the Bluetooth detector system. It is an important parameter for understanding the weekly traffic patterns and identifying the days with higher traffic congestion.

Travel Time Seconds: The Travel Time is the total time calculated from origin reader to destination reader. A series of preprocessing steps have been applied on the dataset and the format of date-time has been converted to

	Record ID	Device Address	Origin Reader Identifier	Destination Reader Identifier	Travel Time Seconds
0	e94781ce761128d4c200358545fbffa6	2609	lamar_45th	lamar_38th	62
1	e73e9874a1cf4e5f822793e66d6fd76c	9395	lamar_45th	lamar_38th	63
2	2f1470b48d514f276911deba6442123b	609	lamar_mlk	lamar_24th	47
3	269960ef63d5606b53c2a47a5e27d081	9798	lamar_24th	lamar_29th	124
4	c552946d20799bcec66ef9d982d9a3ec	7522	lamar_38th	lamar_45th	88
...	...	...	...	...	...

Fig. 4. The data obtained from Bluetooth detectors(first four columns).

	Record ID	Device Address	Origin Reader Identifier	Destination Reader Identifier	Travel Time Seconds
0	e94781ce761128d4c200358545fbffa6	2609	lamar_45th	lamar_38th	62
1	e73e9874a1cf4e5f822793e66d6fd76c	9395	lamar_45th	lamar_38th	63
2	2f1470b48d514f276911deba6442123b	609	lamar_mlk	lamar_24th	47
3	269960ef63d5606b53c2a47a5e27d081	9798	lamar_24th	lamar_29th	124
4	c552946d20799bcec66ef9d982d9a3ec	7522	lamar_38th	lamar_45th	88
...	...	...	...	...	...

Fig. 5. The data obtained from Bluetooth detectors(last four columns).

the actual timestamp here is how it looks like after the dataset.

One hot Encoding: One hot encoding is a concept that is used for such kind of data where there is no ranking provided for the input data and the relationship from one column to another. Consider a data frame with three values A, B, C in it. One hot encoding helps to convert these categotical values into a numerical format. For each category a new binary column is created. In this case, we will have three new columns. In the new columns we assign the value '1' to indicate the presence and the value '0' to indicate the absence. Only one column will have the value '1' and the rest of the columns will have '0's.

Name	Name	A_OneHot	B_OneHot	C_OneHot
A	A	1	0	0
B	B	0	1	0
C	C	0	0	1

The current data frame looks like the this after applying one hot encoding.

We then consider the data in the given format as the input data. In the current research, we applied one hot encoding to the data set with 25748 rows and 9 columns which resulted in 25748 rows and 155 columns. Here is the picture after applying one hot encoding to the same dataset.

Linear regression is a fundamental statistical modeling technique used to understand and predict the relationship between a dependent variable and one or more independent variables. It assumes a linear relationship between the variables, where the dependent variable can be explained by a linear combination of the independent variables. The

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	Origin Location Id_302_51	Origin Location Id_302_Airways	Origin Location Id_302_AlexanderRd	Origin Location Id_302_CraftGoodmanRd	Origin Location Id_302_CraftRd
0	0	0	0	0	0
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	0	0
4	0	0	0	0	0
...	...	...	...	...	...
25743	0	0	0	0	0
25744	0	0	0	0	0
25745	0	0	0	0	0
25746	0	0	0	0	0
25747	0	0	0	0	0

25748 rows × 161 columns

Fig. 6. Image of the data after applying one hot encoding

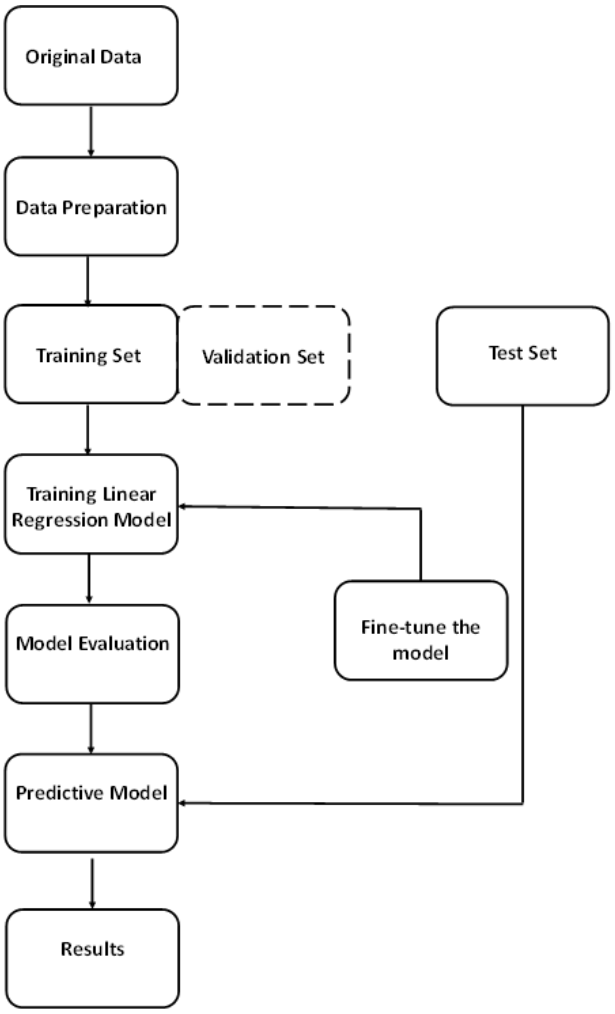


Fig. 7. Data Flow Diagram for Recurrent Neural Networks

goal of linear regression is to estimate the coefficients of the independent variables that minimize the difference between the predicted values and the actual values of the dependent variable. This is done by fitting a straight line that best represents the relationship between the variables. The line is determined by finding the slope (regression coefficient) and the intercept (constant term) that minimize the sum of squared differences between the predicted and actual values.

Once the line is obtained, it can be used to make predictions for new data points. Linear regression is widely used in various fields for tasks and also helps to understand the impact of variables on an outcome. It provides a simple yet powerful tool for modeling and interpreting relationships between variables in a linear context. Considering the current input variables as X_i = (x_1 , x_2 , x_3 , x_4 , x_5 ,....., x_n) where vector 'X' represent the input features and y_i represents the output feature/output variable.

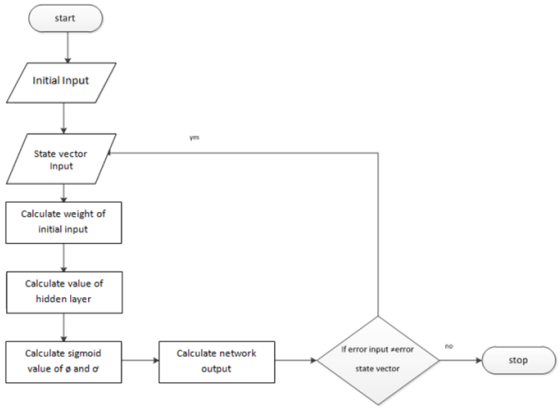


Fig. 8. Linear Regression Flowchart

Recurrent Neural Networks (RNNs) have revolutionized the field of machine learning and have become a driving force behind the advancements in Artificial Intelligence (AI). Inspired by the structure and functionality of biological neural networks, RNNs are computational models composed of interconnected nodes called neurons. These neurons are organized into multiple layers, each responsible for processing and transforming data [17]. RNNs excel at learning complex patterns and relationships from extensive datasets, making them well-suited for many machine learning tasks.

Long Short-Term Memory (LSTM) is a type of recurrent neural network (RNN) that has gained significant attention and success in time series analysis where the travel time prediction problem fits into it. It is designed to overcome the limitations of traditional RNNs by incorporating memory cells and gates that allow the network to capture long-term dependencies in sequential data [18]. These memory cells store and update information over time, enabling LSTMs to effectively process and interpret sequences with varying time lags and dependencies. This architecture enables LSTMs to retain important information for an extended period while selectively forgetting irrelevant details. By leveraging this ability, LSTMs have proven to be highly effective in tasks such as time series forecasting [20].

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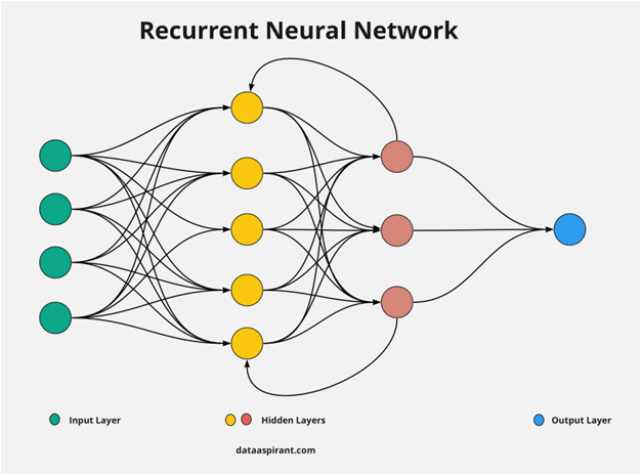


Fig. 9. Recurrent Neural Networks Architecture

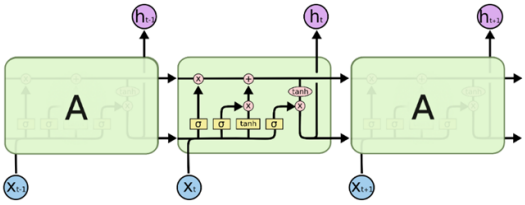


Fig. 10. Long Short-Term Memory Architecture

In this research, we used LSTM networks to model and predict travel time, leveraging their ability to capture and learn from temporal patterns in the data. By incorporating LSTM-based approaches, we aim to enhance the accuracy and reliability of our predictions and contribute to the advancement of predictive modeling techniques for travel time analysis [19].

III. EXPERIMENT

Pandas and numpy libraries are chosen for pre processing of the data. Initially there were some duplicate values that are generated from the data. A duplicate value in this case means that when two vehicles pass two intersections simultaneously then their travel time will be same. Another scenario for a duplicate record in the dataset would be Vehicles moving in the opposite direction with same travel time [15]. These duplicate records will block training the machine learning model in a proper way and will generate squared errors. As a reason, the duplicate records have been removed from the data, and the data frames has been adjusted accordingly.

In this case, Travel time is the Target variable and Speed , Distance(Miles), day, hour, min are the independent variables. As travel time is continuous in this case, we are using Linear Regression and LSTM . The data which we accessed has the weights that are required for neural networks. The key point in this algorithm is, it will try to reduce the mean square error [16].

```
[25] from sklearn.linear_model import LinearRegression
     reg=LinearRegression()
     reg.fit(X_train, y_train)

     ▾ LinearRegression
     LinearRegression()

[26] reg.score(X_test, y_test)

     0.9555167761177804
```

Fig. 11. Long Short-Term Memory Architecture

The Linear Regression model was selected as the baseline algorithm for our regression problem in order to begin building our predictive model. This choice was made for two primary reasons. Firstly, we wanted to gain insights into the individual contributions of the independent variables towards the prediction [21]. Multiple Linear Regression allows us to interpret the coefficients associated with each independent variable, providing a clear understanding of their impact on the outcome. Secondly, we utilized the training and testing data sets as described in the Data partitioning section to evaluate the model's performance.

LSTM is a type of recurrent neural network. LSTM is a specialized variant of RNN that is desgined to address the limitations of traditional RNNs in capturing long-term dependencies.

The reason why we consider LSTM is because a traditional RNNs suffer from the vanishing gradient problem during the back propagation. LSTM will overcome this issue by introducing a complex memory structure. The LSTM cell has additional gating mechanisms that regulate the flow of information, which allows it to selectively retain or forget the information over time. This gating will enable the LSTM to capture the long-term dependencies in sequences more effectively when compared to simple RNNs.

The LSTM incorporates gating mechanisms that allow the model to selectively remember or forget the information at each time step which effectively captures dependencies across the long sequences. The architectural design empowers LSTM to learn and retain information over extended periods, which makes it highly effective when the model is getting trained on time series data.

The key components of an LSTM include the input gate, forget gate, and output gate, which regulate the flow of information and control the memory cell's behavior. This architecture enables LSTMs to retain important information for an extended period while selectively forgetting irrelevant

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```
[29] from keras.models import Sequential
     from keras.layers import Dense, LSTM, Dropout

     # reshape the input data to have 3 dimensions
     X_train = np.reshape(X_train, (X_train.shape[0], 1, X_train.shape[1]))

     # define the LSTM model and add the layers
     model = Sequential()
     model.add(LSTM(units=50, return_sequences=True, input_shape=(1, X_train.shape[2])))
     model.add(LSTM(units=50))
     model.add(Dense(units=1))

     # compiling the model
     model.compile(optimizer='adam', loss='mean_squared_error')

     # training the model
     history = model.fit(X_train, Y_train, batch_size=20, epochs=20, verbose=1)

     loss_values = history.history['loss']
```

Fig. 12. Long Short-Term Memory Implementation

details. By leveraging this ability, LSTMs have proven to be highly effective in tasks such as time series forecasting. In our thesis, we employ LSTM networks to model and predict travel time, leveraging their ability to capture and learn from temporal patterns in the data.

RESULTS

The below is the plot that provides an overview of examining the performance of a linear regression model in the predicting travel time. The scatter plot shows a collection of data points, each representing a specific instance. The red data points correspond to the predicted travel times, while the blue data points represent the actual travel times.

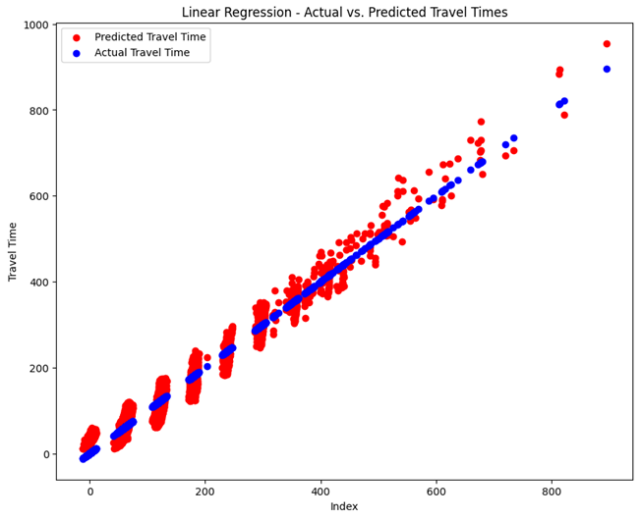


Fig. 13. Linear Regression Analysis

After examining the scatter plot, we observe that some data points fall close to the diagonal line (where predicted and actual values would be equal), indicating accurate predictions. However, there are also points that are scattered away from the diagonal values (actual travel time values), showing discrepancies between the predicted values and actual values.

TABLE I
RMSE VALUES FOR THE MACHINE LEARNING MODEL

Dataset	RMSE
Training Dataset	23.125
Testing Dataset	23.193

Linear Regression model seems to perform reasonably well for certain instances, as evidenced by the data points that are near to the diagonal line.

TABLE II
ACCURACY OF LINEAR REGRESSION MODEL

Model Name	Accuracy
Linear Regression	95.550%

However, there are cases where the model slightly struggles to accurately predict travel times, leading to significant deviations and scattering around the actual values.

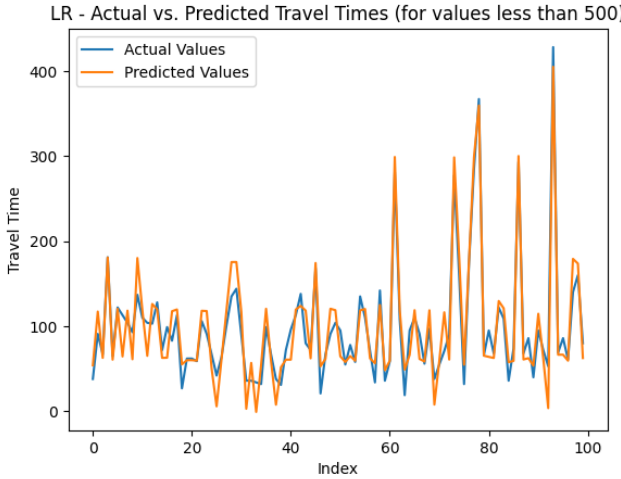


Fig. 14. Linear Regression Analysis

In the LSTM model, there are three main parts: the input layer, the LSTM layer(s), and the output layer. In this code, the first LSTM layer has 50 units and gives output for each step in the sequence. The input data has one time step and a certain number of features. The second LSTM layer also has 50 units but doesn't give sequence output since it's the last layer before the output. The output layer is a Dense layer with one unit, used to predict a single value. This setup works well for predicting continuous values in tasks like regression.

The plotted data points by LSTM algorithm depict a comparison between predicted travel times and actual travel times. The plot shows a general trend of the predicted travel

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times closely aligning with the actual travel times, although there are some instances where there are slight deviations. In most of the cases, the predicted travel times are within a close range of the actual travel times, which clearly indicates a relatively accurate prediction when compared with linear regression.

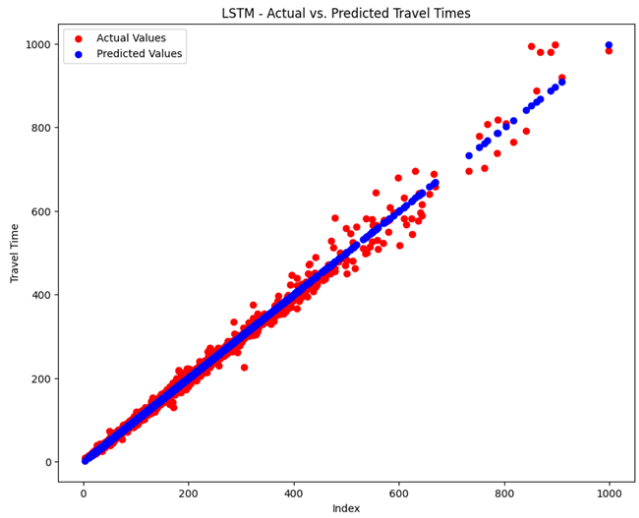


Fig. 15. Actual vs Predicted Time Analysis

However, there are a few outliers where the predicted travel times significantly differ from the actual travel times. These outliers also suggest instances where the prediction model may have encountered challenges or uncertainties in accurately estimating the travel time. Overall, the plot demonstrates a reasonable level of prediction accuracy, with the majority of the data points falling close to the line of equality, indicating a reliable prediction model for travel times.

TABLE III
RMSE VALUES FOR THE DATASETS

Dataset	RMSE
Training Dataset	11.042
Testing Dataset	11.952

The LSTM algorithm in our experiment showed impressive performance, with RMSE values of 11.04 for the training set and 11.95 for the testing set. These values indicate that the algorithm is making very accurate predictions, When the RMSE values are similar for both the training and testing sets, it means that the algorithm is fitting well to the data and able to make reliable predictions even on new, unseen data. The LSTM model impressively demonstrates its capability to continuously enhance and fine-tune its predictions, as clearly illustrated by the remarkable decrease in loss throughout the training process.

TABLE IV
ACCURACY OF LSTM MODEL

Model Name	Accuracy
Long Short Term Memory	97.862%

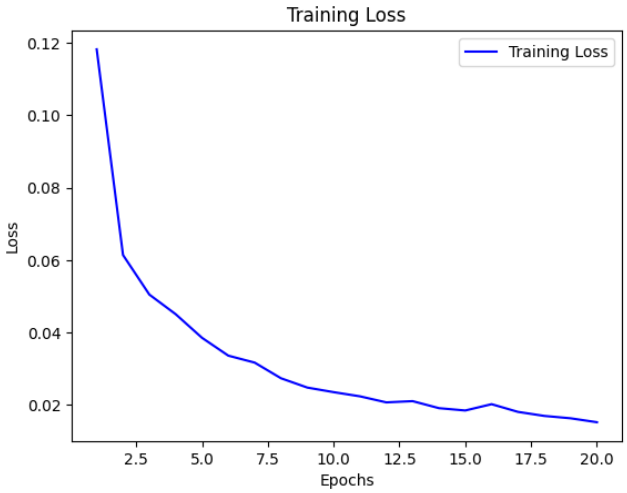


Fig. 16. Optimization of Loss Function

Starting from an initial loss of 0.1183 in the first epoch, the LSTM algorithm steadily improves its performance, achieving an impressive loss of 0.0152 in the final epoch.

This substantial reduction in loss signifies the algorithm's ability to learn and capture the complex patterns and trends within the data.

This paper introduces a real-time travel time estimation model utilizing Bluetooth sensor data. The primary objective is to accurately estimate travel time, enabling its practical application in congestion management and similar real-life scenarios. To ensure reliable analysis results, a set of procedures and algorithms based on traffic flow principles were introduced. Furthermore, novel techniques like one-hot encoding were employed to preprocess the data, making it suitable for training and testing various models.

LIMITATIONS

The research paper offers a novel approach to travel time prediction using machine learning models and real-time Bluetooth sensor data. However, there are a few limitations that needs to considered. Firstly, the study is confined to highways equipped with Bluetooth sensors. Secondly, the reliance on Bluetooth sensor data may not be feasible in the regions that lack such infrastructure. Furthermore,the Bluetooth sensors are designed to operate and capture the information from 1 foot to a distance of about 300 feet, beyond this range, the sensors will become incapable and will fail to detect the Bluetooth information.

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FUTURE WORK

In future work, the focus will be on enhancing the generalization capability of the deep learning model by incorporating a larger volume of required data. This includes integrating additional traffic parameters that influence traffic conditions, such as stop signs, construction zones, and speed limits on road segments. Once the deep learning model achieves optimal accuracy on the existing road network, the next step involves expanding the model to encompass other links within the road network while incorporating the newly added parameters. By making this information available, the deep learning model can provide real-time travel time estimates with the utmost accuracy.

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